Machine Intelligence 1 Tutorial

13 Reinforcement Learning Pt 1/2 (Value estimation)

Fachgebiet Neuronale Informationsverarbeitung (NI)

WiSe 25/26

The Reinforcement Learning (RL) setting

Markov Decision Process (MDP)

Model-based evaluation of the policy

References

The Reinforcement Learning (RL) setting

Reminder:

- Data in supervised learning: $\left\{\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N, \underline{\mathbf{y}}_T^{(p)} \in \mathbb{R}^M\right\}_{\alpha=1}^{p}$
- Data in **un**supervised learning: $\left\{\underline{\mathbf{x}}^{(\alpha)} \in \mathbb{R}^N\right\}_{\alpha=1}^{p}$

Data in Reinforcement Learning:

A sequence of random states $\underline{\mathbf{x}}$

resulting from taking actions $\underline{\mathbf{a}}$ over time

which result in some reward r.

The Reinforcement Learning (RL) setting

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  e.g. $\mathcal{X} := \{\underline{\mathbf{x}}_1, \ldots, \underline{\mathbf{x}}_S\} \subset \{0,1\}^S$ (1-out-S encoding)

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The sequence we observe becomes:

$$\left\{\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}, r^{(t)}\right\}_{t=0}^{p} = \left(\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)}, r^{(0)}\right), \ldots, \left(\underline{\mathbf{x}}^{(p)}, \underline{\mathbf{a}}^{(p)}, r^{(p)}\right)$$

Markov property

The probability of transitioning from state $\underline{\mathbf{x}}_i$ to $\underline{\mathbf{x}}_j$ only depends on state $\underline{\mathbf{x}}_i$ and the action $\underline{\mathbf{a}}_k$ that was taken. It does not depend on earlier history.

This leads to the following transition model (transition "matrix" or "volume"):

$$P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)$$

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- is a stationary distribution

$$P(\underline{\mathbf{x}}_j^{(t+1+\tau)} | \underline{\mathbf{x}}_i^{(t+\tau)}, \underline{\mathbf{a}}_k^{(t+\tau)}) = P(\underline{\mathbf{x}}_j^{(t+1)} | \underline{\mathbf{x}}_i^{(t)}, \underline{\mathbf{a}}_k^{(t)}), \quad \tau \in \{0, 1, \ldots\}$$

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and where

$$P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \geq 0 \quad \forall i, j, k$$

and $\sum_j P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) = 1 \quad \forall i, k$ i.e. every "row" sums to 1

Properties of the transition model

Q: *What would* $\sum_{\substack{j \\ j \neq i}} P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) = 0 \;\; \forall \, k$ *imply?*

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transition model (transition matrix):

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Markov chain

a sequence of random states $\underline{\mathbf{x}}$ with a *Markov property*.

Policy

Representation of the agent[a]. It measures the distribution over actions given states.

- map states to actions, i.e. the probability of doing $\underline{\mathbf{a}}_k$ when in $\underline{\mathbf{x}}_i$:

$$\text{policy } \pi := \pi(\underline{\mathbf{a}}_k | \underline{\mathbf{x}}_i)$$

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- the mapping can be stochastic or deterministic
- the policy is stationary. That the policy changes during the training process does not contradict this. It is regarded as stationary because once trained and deployed, the way the agent decides does not change over time.

[a] Without a policy we would only have a Markov Reward Process. Having a policy (wanting to find a policy) turns the Markov Reward Process into a Markov Decision Process.

Markov Decision Process (MDP)

A formalized description of the environment in RL[1]

[1] Recommended reading (Sutton et al., 1998) also video lectures by David Silver

Markov Decision Process (MDP)

Assumption of MDP

The environment is fully observable. There is a complete characterization of what's going on and where everything is.

The other case would be *partially observable* MDP (POMDP)

Markov Decision Process (MDP)

What do we want to do?

Maximize, not just immediate rewards, but the *cumulative sum of rewards* or "return".

$$return(t) := \sum_{k=0}^{\infty} \gamma^k r(\underline{\mathbf{x}}^{(t+k+1)}, \underline{\mathbf{a}}^{(t+k+1)}) = \sum_{k=0}^{\infty} \gamma^k r^{(t+k+1)},$$

where $\gamma \in [0, 1)$ is referred to as the *discount factor*.

About the *discount factor* γ:

- It is possible for γ to be $= 1$.

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  - emphasis on long-term returns, indifference to delay.

The discount factor

$$return(t) := \sum_{k=0}^{\infty} \gamma^k r^{(t+k+1)}, \quad \text{where } \gamma \in [0, 1)$$

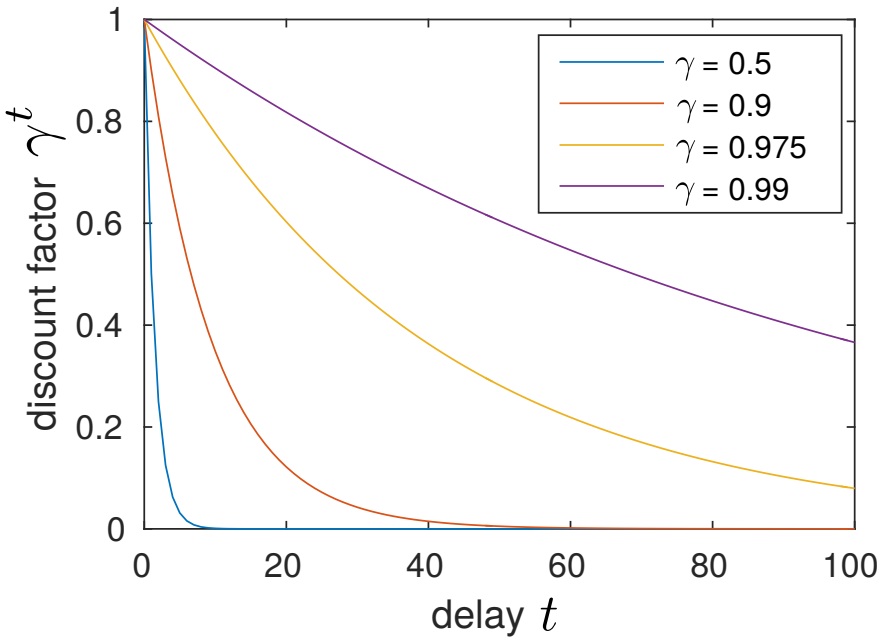

Figure: modulating returns with different discount factors

Example: big reward further away vs. closer smaller reward

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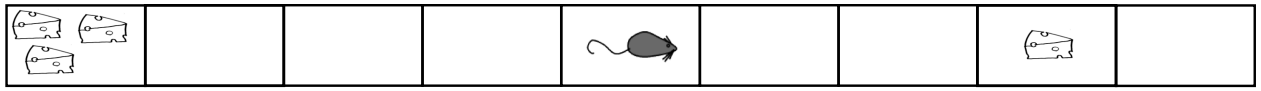

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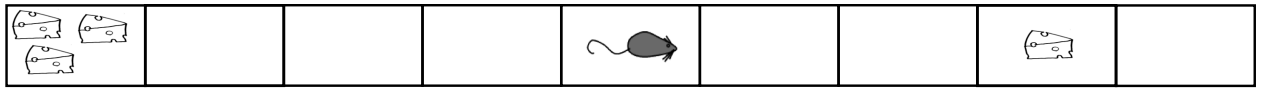

Comparing two strategies under different discount factors:

- $\longrightarrow, \longrightarrow, \longrightarrow, \ldots$
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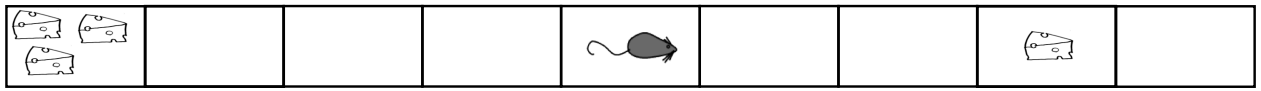

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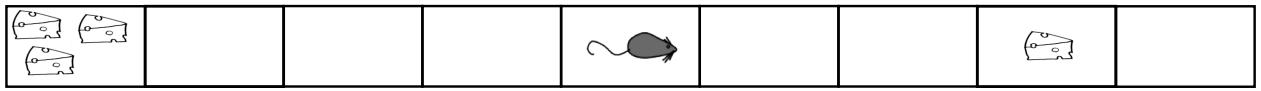

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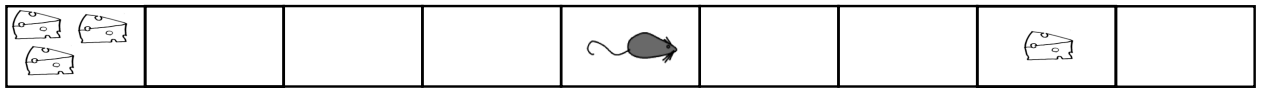

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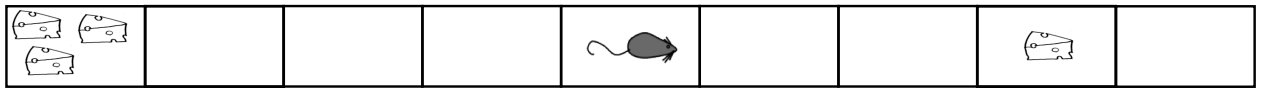

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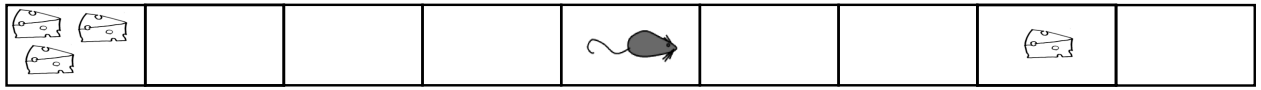

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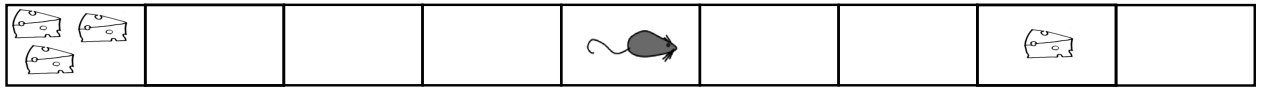

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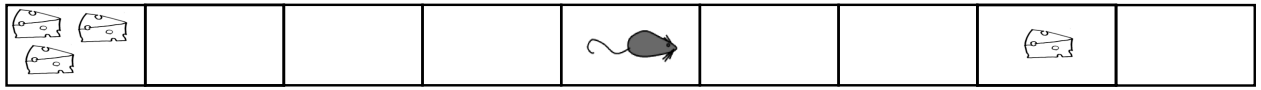

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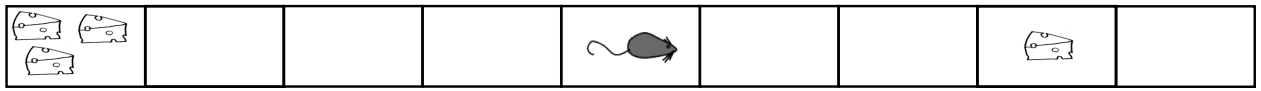

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  γ=0.2: $return(0)=$

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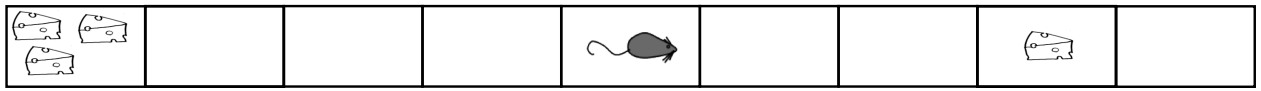

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  γ=0.2: $return(0)=1 \cdot 0 + 0.2 \cdot 0 + 0.2^2 \cdot 1 + \ldots \geq \mathbf{0.04}$

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  γ=0.2: $return(0)=1 \cdot 0 + 0.2 \cdot 0 + 0.2^2 \cdot 0 + 0.2^3 \cdot 3 + \ldots \geq 0.024$

Q: *Why do we discount future rewards? What criteria go into choosing γ?*

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- Uncertainty of the future
- Imperfections in the model (we don't trust its decisions completely)
- mathematically convenient, the sum does not explode
- avoids ∞ returns due to possible cycles in our MDP
- behavioral arguments (animals and humans seem to apply a similar discounting)
  "A bird in the hand is worth two in the bush"
- financially realistic (accounts for inflation)

The value function

The (state) value function $\widehat{=}$ the expected sum of discounted future rewards

Q: *What does the value function represent?*

- The long-term value of some state $\underline{\mathbf{x}}$,
- the expected return from starting in $\underline{\mathbf{x}}^{(0)}$,
- we have a preference for high expectations and MDPs maximize this quantity

The value function

The value function $\widehat{=}$ the expected sum of discounted future rewards
For a fixed policy π:

- a **value function** measures the quality of a policy π in state $\underline{\mathbf{x}}^{(0)}$
  - $V^{\pi}(\underline{\mathbf{x}}^{(0)})$ is the *expected* *reward*

$$V^{\pi}(\underline{\mathbf{x}}^{(0)}) = \mathbb{E}\left[r(\underline{\mathbf{x}}^{(0)}, \underline{\mathbf{a}}^{(0)}) \,\middle|\, \underline{\mathbf{a}}^{(0)} \sim \pi(\cdot \mid \underline{\mathbf{x}}^{(0)}) \right]$$

The value function

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$$V^{\pi}(\underline{\mathbf{x}}^{(0)}) = \mathbb{E}\left[\sum_{t=0}^{H} r(\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}) \;\middle|\; \begin{array}{c} \underline{\mathbf{a}}^{(t)} \sim \pi(\cdot \mid \underline{\mathbf{x}}^{(t)}) \\ \underline{\mathbf{x}}^{(t+1)} \sim P(\cdot \mid \underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}) \end{array}\right]$$

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The value function $\widehat{=}$ the expected sum of discounted future rewards

For a fixed policy π:

- a **value function** measures the quality of a policy π in state $\underline{\mathbf{x}}^{(0)}$
  - $V^{\pi}(\underline{\mathbf{x}}^{(0)})$ is the *expected infinite sum of discounted future rewards*

$$V^{\pi}(\underline{\mathbf{x}}^{(0)}) \;=\; \mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^{t}\, r(\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}) \;\middle|\; \begin{array}{c} \underline{\mathbf{a}}^{(t)} \sim \pi(\cdot \,|\, \underline{\mathbf{x}}^{(t)}) \\ \underline{\mathbf{x}}^{(t+1)} \sim P(\cdot \,|\, \underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}) \end{array}\right], \quad \gamma \in [0,1)\,.$$

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In other textbooks you may find:

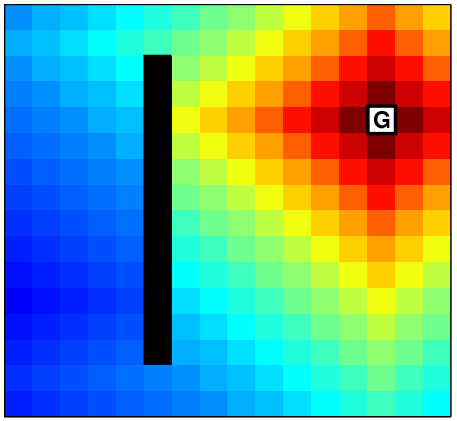
$$V^{\pi}(\underline{\mathbf{x}}_i) = \overbrace{V_i^{\pi}}^{\text{shorthand}} = \mathbb{E}\left[\sum_{t=0}^{\infty}\gamma^{t} r(\underline{\mathbf{x}}^{(t)},\underline{\mathbf{a}}^{(t)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right],\; i = 1,\ldots,S$$

$\mathbb{E}[\cdot]$ is w.r.t $\{\underline{\mathbf{x}}^{(t)},\underline{\mathbf{a}}^{(t)}\}$

The value function

The value function $\widehat{=}$ the expected sum of discounted future rewards
For a fixed policy π:

value V^π



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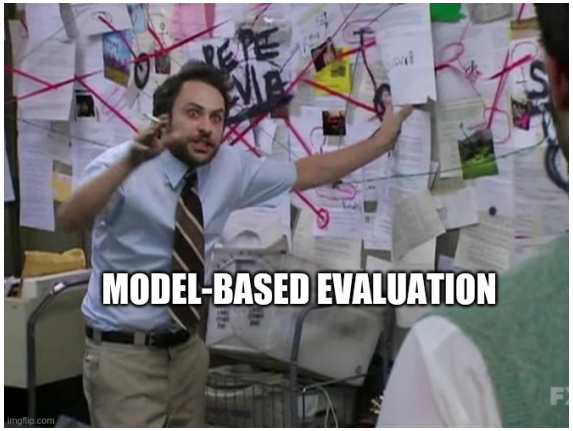
$\mathbb{E}[\cdot]$ is w.r.t $\{\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)}\}$

Model-based evaluation of the policy

Model-based $\widehat{=}$ offline-planning.
We are told everything about the environment,
so we can simulate "looking ahead" and
determine if one policy is better than another.



The Bellman equation

The Bellman equation relates the value of a state with the values of its successor states, given some policy π.
The policy is fixed beforehand, i.e. V "follows" the policy π.

The Bellman equation

$$
\begin{aligned}
V_i^\pi &= \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t \overbrace{r(\underline{\mathbf{x}}^{(t)}, \underline{\mathbf{a}}^{(t)})}^{\substack{\text{shorthand} \\ =:r^{(t)}}} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big], \; i = 1, \dots, S \\
&= \mathbb{E}\Big[\sum_{t=0}^{\infty} \gamma^t r^{(t)} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \\
&= \mathbb{E}\Big[\gamma^0 \underbrace{r^{(0)}}_{\substack{\text{immediate} \\ \text{reward}}} + \gamma^1 r^{(1)} + \gamma^2 r^{(2)} + \dots \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \\
&= \mathbb{E}\Big[r^{(0)} + \underbrace{\gamma\big(r^{(1)} + \gamma r^{(2)} + \dots\big)}_{\substack{\text{discounted value of} \\ \text{successor state}}} \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big] \\
&= \mathbb{E}\Big[r^{(0)} + \gamma V^\pi(\underline{\mathbf{x}}^{(1)}) \;\Big|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\Big]
\end{aligned}
$$

The Bellman equation

$$V_i^\pi = \mathbb{E}\left[r^{(0)} + \gamma V^\pi(\underline{\mathbf{x}}^{(1)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right]$$

split "expectation of sum" into "sum of expectations":

$$= \mathbb{E}\left[r^{(0)} \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right] + \mathbb{E}\left[\gamma V^\pi(\underline{\mathbf{x}}^{(1)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right]$$

$$= \mathbb{E}\left[r^{(0)} \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right] + \gamma\,\mathbb{E}\left[V^\pi(\underline{\mathbf{x}}^{(1)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right]$$

The Bellman equation (cont'd)

$$V_i^\pi = \mathbb{E}\left[r^{(0)} \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right] + \gamma\,\mathbb{E}\left[V^\pi(\underline{\mathbf{x}}^{(1)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right]$$

Recall:

states $\underline{\mathbf{x}} \in \{\underline{\mathbf{x}}_1, \ldots, \underline{\mathbf{x}}_S\}$, actions $\underline{\mathbf{a}} \in \{\underline{\mathbf{a}}_1, \ldots, \underline{\mathbf{a}}_A\}$,
reward function $r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)$,
transition model $P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)$ and policy $\pi(\underline{\mathbf{a}}_k | \underline{\mathbf{x}}_i)$

Q: *What does the* $\mathbb{E}[\cdot]$ *boil down to?*

The Bellman equation (cont'd)

$$V_i^\pi = \mathbb{E}\left[r^{(0)} \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right] + \gamma\,\mathbb{E}\left[V^\pi(\underline{\mathbf{x}}^{(1)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right]$$

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Q: *What does the* $\mathbb{E}[\cdot]$ *boil down to?*

$$V_i^\pi = \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \mid \underline{\mathbf{x}}_i) r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\text{"controlled" reward function (immediate reward)}=:r_i^\pi} +$$

The Bellman equation (cont'd)

$$V_i^\pi = \mathbb{E}\left[r^{(0)} \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right] + \gamma\,\mathbb{E}\left[V^\pi(\underline{\mathbf{x}}^{(1)}) \;\middle|\; \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right]$$

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$$V_i^\pi = \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \mid \underline{\mathbf{x}}_i) r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\substack{\text{“controlled” reward function} \\ \text{(immediate reward)}=:r_i^\pi}} + \underbrace{\gamma \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \mid \underline{\mathbf{x}}_i) \sum_{j=1}^{S} P(\underline{\mathbf{x}}_j \mid \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \overbrace{V_j^\pi}^{=V^\pi(\underline{\mathbf{x}}_j)}}_{\text{expected discounted value of successor state}}$$

$$=$$

The Bellman equation (cont'd)

$$V_i^\pi = \mathbb{E}\left[r^{(0)} \,\middle|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right] + \gamma\,\mathbb{E}\left[V^\pi(\underline{\mathbf{x}}^{(1)}) \,\middle|\, \underline{\mathbf{x}}^{(0)} := \underline{\mathbf{x}}_i\right]$$

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$$V_i^\pi = \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\text{"controlled" reward function (immediate reward)}=:r_i^\pi} + \underbrace{\gamma \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) \sum_{j=1}^{S} P(\underline{\mathbf{x}}_j \,|\, \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \overbrace{V_j^\pi}^{=V^\pi(\underline{\mathbf{x}}_j)}}_{\text{expected discounted value of successor state}}$$

$$= \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\text{"controlled" reward function } r_i^\pi} + \gamma \sum_{j=1}^{S} \underbrace{\sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i) P(\underline{\mathbf{x}}_j \,|\, \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k)}_{\text{"controlled" transition model } P_{ij}^\pi} V_j^\pi$$

$$\underline{\mathbf{v}}^\pi = \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi$$

$$\underline{\mathbf{v}}^{\pi} = \underline{\mathbf{r}}^{\pi} + \gamma \underline{\mathbf{P}}^{\pi} \underline{\mathbf{v}}^{\pi} \;=:\; \overbrace{\hat{B}^{\pi}[\underline{\mathbf{v}}^{\pi}]}^{\text{the Bellman operator}} \;,$$

where $\underline{\mathbf{v}}^{\pi} \in \mathbb{R}^{S}$ is a vector containing all values $V^{\pi}(\underline{\mathbf{x}}_i)\ \forall i$,

$$\text{with} \underbrace{\begin{cases} r_i^{\pi} := \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i)\, r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \\ P_{ij}^{\pi} := \sum_{k=1}^{A} \pi(\underline{\mathbf{a}}_k \,|\, \underline{\mathbf{x}}_i)\, P(\underline{\mathbf{x}}_j | \underline{\mathbf{x}}_i, \underline{\mathbf{a}}_k) \end{cases}}_{\text{“controlled” models } \underline{\mathbf{r}}^{\pi} \in \mathbb{R}^{S} \text{ and } \underline{\mathbf{P}}^{\pi} \in \mathbb{R}^{S \times S}}$$

Finding $\underline{\mathbf{v}}^{\pi}$

$$\underline{\mathbf{v}}^{\pi} = \underline{\mathbf{r}}^{\pi} + \gamma \underline{\mathbf{P}}^{\pi} \underline{\mathbf{v}}^{\pi}$$

There are two approaches to finding $\underline{\mathbf{v}}^{\pi}$:

1 - analytically

2 - through fixed-point iteration

Finding $\underline{\mathbf{v}}^\pi$: Analytical solution of the Bellman equation

Essentially solve for $\underline{\mathbf{v}}^\pi$ (get <u>all</u> $\underline{\mathbf{v}}^\pi$ on one side):

$$\begin{aligned}
\underline{\mathbf{v}}^\pi &= \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi \\
\underline{\mathbf{I}}\ \underline{\mathbf{v}}^\pi &= \underline{\mathbf{r}}^\pi + \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi \\
\underline{\mathbf{I}}\ \underline{\mathbf{v}}^\pi - \gamma \underline{\mathbf{P}}^\pi \underline{\mathbf{v}}^\pi &= \underline{\mathbf{r}}^\pi \\
\left(\underline{\mathbf{I}} - \gamma \underline{\mathbf{P}}^\pi\right) \underline{\mathbf{v}}^\pi &= \underline{\mathbf{r}}^\pi \\
\underline{\mathbf{v}}^\pi &= \left(\underline{\mathbf{I}} - \gamma \underline{\mathbf{P}}^\pi\right)^{-1} \underline{\mathbf{r}}^\pi
\end{aligned}$$

Q: *When is the term* $\left(\underline{\mathbf{I}} - \gamma \underline{\mathbf{P}}^\pi\right)$ *invertible?*

1. $|\lambda_k| \leq 1$ for all eigenvalues λ_k of transition matrix $\underline{\mathbf{P}}^\pi$
2. $\gamma < 1$

$\Rightarrow$ it is always invertible

Finding $\underline{\mathbf{v}}^{\pi}$: Analytical solution of the Bellman equation

$$\underline{\mathbf{v}}^{\pi} = \left(\underline{\mathbf{I}} - \gamma \underline{\mathbf{P}}^{\pi}\right)^{-1} \underline{\mathbf{r}}^{\pi}$$

Q: *Any disadvantages to the analytical solution?*

Finding $\underline{\mathbf{v}}^{\pi}$: Analytical solution of the Bellman equation

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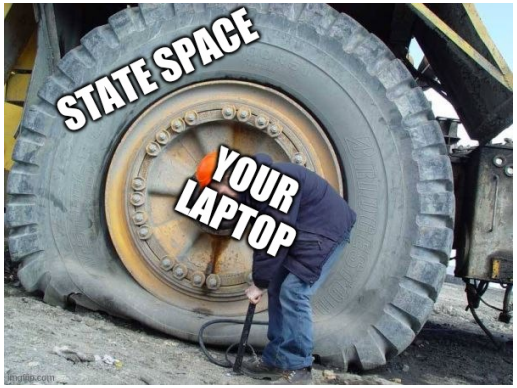
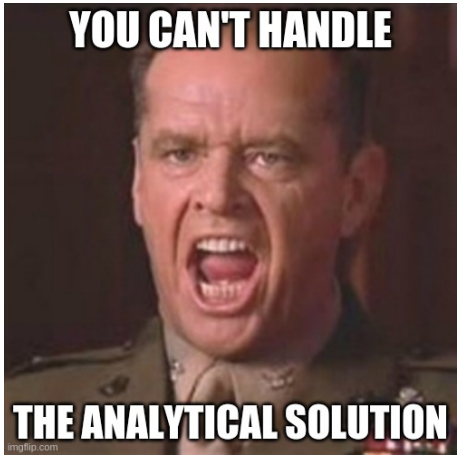
Q: *Any disadvantages to the analytical solution?*



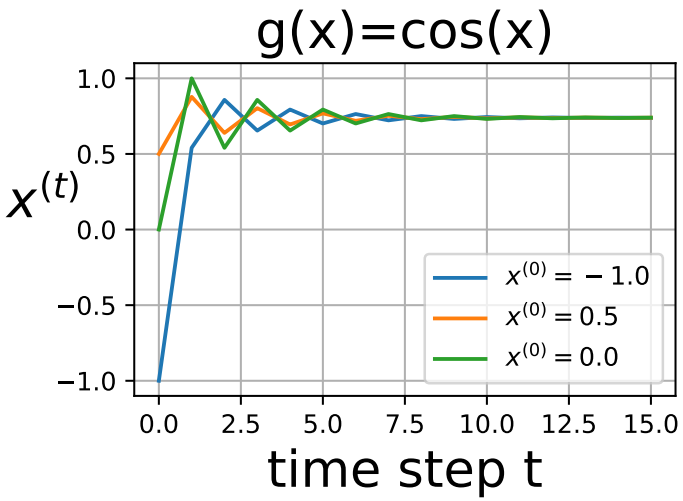


- It could be very costly for very large state spaces.

Finding $\underline{\mathbf{v}}^{\pi}$: Value iteration

Value iteration: a fixed-point iteration for finding $\underline{\mathbf{v}}^{\pi}$

- A fixed point: $x = g(x) = g(g(\ldots g(x)\ldots))$
  Example:
  $$g(x) = x^2 - 3x + 4$$
  $$g(2) = 2^2 - 3 \cdot 2 + 4 = 4 - 6 + 4 = 2$$
- Fixed point iteration:
  $$x^{(t+1)} = g(x^{(t)}), t = 0, 1, 2, \ldots$$



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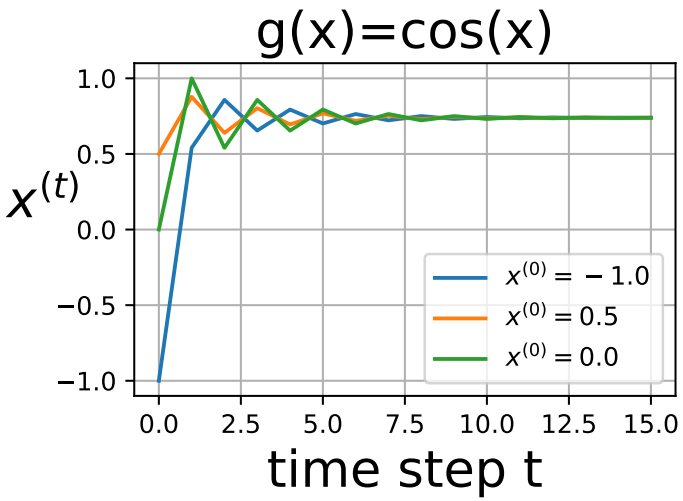

Finding $\underline{\mathbf{v}}^{\pi}$: Value iteration

Value iteration: a fixed-point iteration for finding $\underline{\mathbf{v}}^{\pi}$

$$\underline{\tilde{\mathbf{v}}}^{\pi(t+1)} = \underline{\mathbf{r}}^{\pi} + \gamma \underline{\mathbf{P}}^{\pi} \underline{\tilde{\mathbf{v}}}^{\pi(t)}$$

- The "~" is to denote that it has not necessarily converged at time step t
- initialize $\underline{\tilde{\mathbf{v}}}^{\pi(0)}$ with some state $\underline{\tilde{\mathbf{v}}}^{\pi(0)} \in \mathbb{R}^S$
- value iteration converges in the limit $t \to \infty$
- The speed of convergence depends on γ

Q: *How does convergence depend on γ?*

Finding $\underline{\mathbf{v}}^{\pi}$: Value iteration

Value iteration: a fixed-point iteration for finding $\underline{\mathbf{v}}^{\pi}$

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- The speed of convergence depends on γ

Q: *How does convergence depend on γ?*

- small γ: fast convergence
- $\gamma \to 1$: slow convergence, slower than the analytical solution

Finding $\underline{\mathbf{v}}^{\pi}$: Value iteration

$$\underline{\tilde{\mathbf{v}}}^{\pi(t+1)} = \underline{\mathbf{r}}^{\pi} + \gamma \underline{\mathbf{P}}^{\pi} \underline{\tilde{\mathbf{v}}}^{\pi(t)}$$

- We start *value iteration* assuming $\underline{\tilde{\mathbf{v}}}^{\pi(0)}$ is the optimal value function
- Use the one-step *look-ahead* (i.e. next step) to work our way *backwards*

$$\hat{B}^{\pi}[\underline{\mathbf{v}}^{\pi}] := \underline{\mathbf{r}}^{\pi} + \gamma \underline{\mathbf{P}}^{\pi} \underbrace{\underline{\mathbf{v}}^{\pi}}_{\text{look-ahead}}$$

Analytical solution vs. value iteration

- complexity:
  - Analytical solution: $O(S^3)$
  - value iteration: $O(S^2)$
- intermediate solutions of *value iteration* don't necessarily belong to valid policies. We have to wait for convergence.

Finding the optimal policy

Comparing two policies

My policy is better than yours

"better" or
equally "good"

$$\pi \geq \pi' \quad \textbf{iff} \quad V^{\pi}(\underline{\mathbf{x}}) \geq V^{\pi'}(\underline{\mathbf{x}}), \quad \forall\, \underline{\mathbf{x}} \in \mathcal{X}$$

Finding the optimal policy

Multiple *optimal policies* can exist. A policy is optimal if it maximizes the state value function:

$$V^*(\underline{\mathbf{x}}) = \max_{\pi} V^{\pi}(\underline{\mathbf{x}}), \quad \forall\, \underline{\mathbf{x}} \in \mathcal{X}$$

Optimal Policy

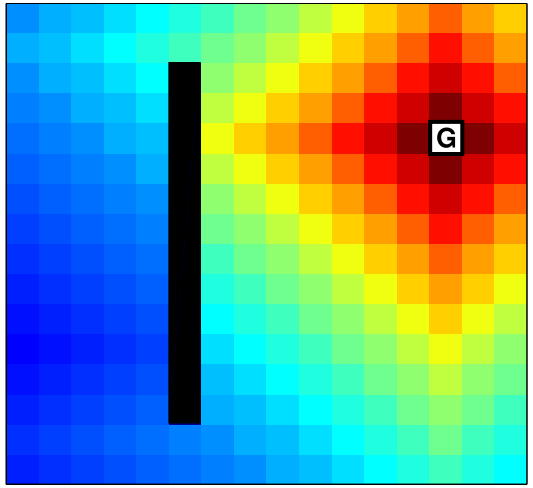
For all optimal policies π^*:

$$V^{\pi^*}(\underline{\mathbf{x}}) = V^*(\underline{\mathbf{x}})$$

A policy π that maximizes the value function and yields $\underline{\mathbf{v}}^*$ is an optimal policy π^*.

Extracting the policy from an optimal value function

value V^{π}



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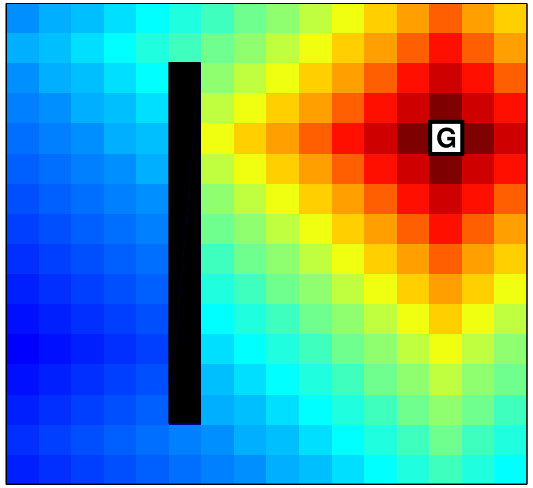
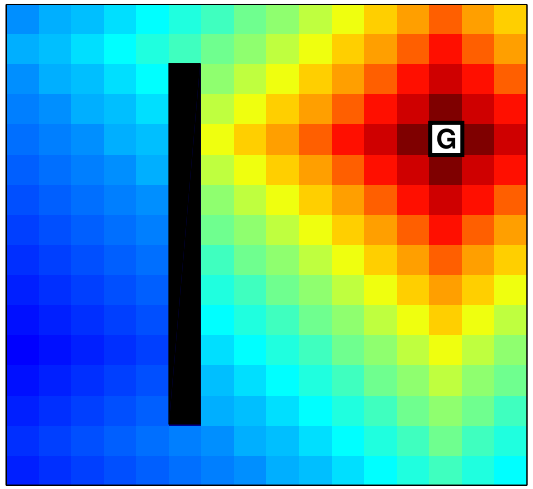


$$\pi^*(\underline{\mathbf{a}}_k|\underline{\mathbf{x}}_i) = \underset{\underline{\mathbf{a}}}{\operatorname{argmax}} \left(r(\underline{\mathbf{x}}_i, \underline{\mathbf{a}}) + \gamma \sum_{j=1}^{S} P(\underline{\mathbf{x}}_j|\underline{\mathbf{x}}_i, \underline{\mathbf{a}}) V^*(\underline{\mathbf{x}}_j) \right)$$

Extracting the policy from an optimal value function

value V^π



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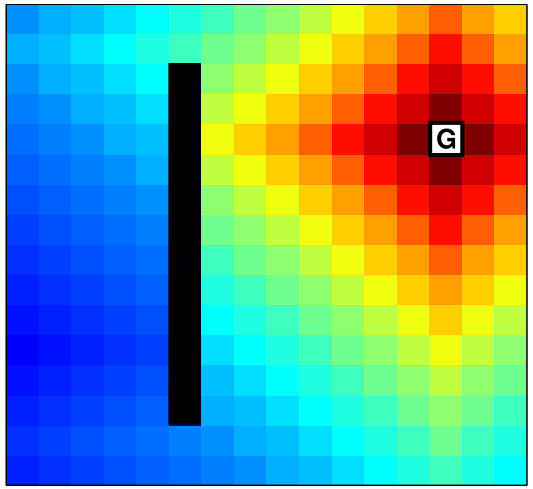
Q: *What is going on here?*

1. When at $\underline{\mathbf{x}}_i$, look at the actions $\underline{\mathbf{a}}$ that lead to all possible future states $\underline{\mathbf{x}}_j$.
2. Select action $\underline{\mathbf{a}}_k$ that maximizes the immediate reward + discounted successive value.

Extracting the policy from an optimal value function

value V^π



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Q: *What are the implications of selecting the optimal policy this way?*

- The above does not scale well for large state spaces.
- The resulting π^* is a *greedy* policy, because it greedily selects using only the value function at hand.

References I

Richard S Sutton, Andrew G Barto, et al. *Introduction to reinforcement learning*, volume 135. MIT press Cambridge, 1998.